
Event Generation with Parallel Langevin Sampling and Learned Stein Diagnostics

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Efficient event generation is a major computational challenge for precision collider
2 simulations, especially for high-multiplicity final states where matrix-element
3 evaluations are expensive and rejection-sampling efficiencies are low. We study
4 an alternative approach based on many parallel underdamped Langevin chains,
5 retaining one terminal state from each chain to obtain unweighted events while
6 avoiding within-chain autocorrelation. A learned Stein discrepancy is used as a
7 convergence diagnostic, providing a data-driven estimate of the relaxation time.
8 We apply the method to tree-level $u\bar{u} \rightarrow Z + n g$ event generation and find that
9 relaxation requires only a modest number of exact-target Langevin steps, with
10 mild growth over the multiplicities studied. Finally, we show that simple neural-
11 network surrogate initialization can substantially reduce the required number of
12 exact matrix-element and gradient evaluations.

13 1 Introduction

14 Monte Carlo event simulation is central to particle physics, connecting theoretical predictions with
15 experimental measurements. The high-luminosity upgrade to the Large Hadron Collider (LHC) [1]
16 and its potential successors [2, 3] will demand unprecedented precision and correspondingly large
17 simulated samples, whose production is expected to become a dominant computational bottleneck [4–
18 8].

19 The most expensive component of the simulator chain [9] is the sampling of the hard proton-proton
20 collision differential cross section

$$d\sigma_{pp \rightarrow X_n} = \frac{1}{\mathcal{F}} \sum_{i,j} f_i(x_1) f_j(x_2) |M_{ij \rightarrow X_n}(\Phi_n)|^2 dx_1 dx_2 d\Phi_n, \quad (1)$$

21 where the sum runs over the constituents of the proton (quarks, antiquarks and gluons), f are their
22 corresponding parton density functions, $|M_{ij \rightarrow X_n}|^2$ is the squared matrix element, and $\mathcal{F}$ is the
23 incoming partonic flux factor. The integration measure is given by the two partonic momentum
24 fractions $dx_{1,2} \in [0, 1]$ and the n -body phase-space measure $d\Phi_n$, which integrates over the final-
25 state momenta while ensuring momentum conservation. The total sampling dimensionality of this
26 object is $3n - 2$.

27 For large multiplicities n and for high theoretical precision, the evaluation of the matrix element
28 can be prohibitively expensive. The method used to sample Eq. (1) must therefore be as efficient as
29 possible in terms of evaluations, even if this comes at increased algorithmic cost. Current methods are
30 still largely based on the VEGAS algorithm [10, 11] with many physics-inspired improvements [12–
31 19]. These methods all rely on an adaptive approximation of an unnormalized target density $\tilde{p}(x)$ (the
32 differential cross section) by some density $q(x)$ that is easily sampled. One can then sample from

33 $q(x)$ and assign weights $w(x) = \tilde{p}(x)/q(x)$. To obtain unweighted samples, rejection sampling must
 34 be performed with acceptance probability

$$p_{\text{acc}}(x) = \frac{w(x)}{\max_x w(x)}. \quad (2)$$

35 The unweighting efficiency $\epsilon = \mathbb{E}_{x \sim q} [p_{\text{acc}}(x)]$ then determines the required number of matrix
 36 element evaluations per generated sample. Eq. 2 is very sensitive to the value of $\max_x w(x)$, which
 37 may originate from tiny regions of phase space. Thus, to improve efficiency, one may make use of a
 38 reduced maximum and leave some undesirable weighted events [20]. Nevertheless, for increasing
 39 multiplicities, the unweighting efficiency can still degrade by several orders of magnitude [21].

40 Recent work has used modern machine-learning (ML) architectures such as normalizing flows in
 41 place of VEGAS to improve efficiencies [22–29]. These approaches are however still ultimately
 42 based on rejection sampling and thus inherit its disadvantages. In this work, we propose a different
 43 approach based on parallelized Markov chain Monte Carlo (MCMC) supplemented by an ML-based
 44 convergence diagnostic. MCMC methods have previously been explored for hard-process phase-
 45 space sampling and event generation [30–33]. Our method places particular emphasis on avoiding
 46 within-chain autocorrelations and on reaching MCMC stationarity, i.e. samples being representative
 47 of $\tilde{p}(x)$. Furthermore, we investigate how neural network surrogates of the target distribution can
 48 further reduce the required number of matrix-element evaluations [34–36].¹

49 2 Method

50 We now describe our sampling construction, targeting an unnormalized density $\tilde{p}(x)$ on the unit
 51 hypercube $x \in [0, 1]^d$. To avoid within-chain autocorrelation, we evolve many Markov chains in
 52 parallel on hardware accelerators and retain only the terminal state from each chain. The learned
 53 Stein discrepancy introduced below is used to determine the relaxation time t_{rel} , i.e. the number of
 54 MCMC steps after which the ensemble is representative of the target.

55 2.1 Underdamped Langevin Algorithm

56 Our MCMC framework is designed for rapid convergence under two practical constraints: the sampler
 57 should be straightforward to run over many parallel chains, and it should use expensive target-density
 58 and gradient evaluations sparingly. Sophisticated Hamiltonian Monte Carlo (HMC) methods [37]
 59 like NUTS [38] implement adaptive trajectory lengths which leads to complex control flow and thus
 60 difficult parallelizability. Moreover, HMC typically requires several intermediate gradient evaluations
 61 before a single Metropolis-Hastings (MH) correction is applied. We therefore use an Underdamped
 62 Langevin Algorithm (ULA) [39, 40], which retains momentum between steps while applying an
 63 MH correction after each proposal. Each parallel chain evolves a position–momentum pair using a
 64 leapfrog proposal, a MH correction, partial momentum refreshment, and adaptive step sizes. The
 65 complete update and adaptation scheme is given in Appendix A.

66 2.2 Mirror Langevin Dynamics

67 After the application of a mapping, the phase-space coordinates used for the cross-section integrand
 68 live on the bounded domain $x \in [0, 1]^d$, while ULA is most naturally formulated on $\mathbb{R}^d$. We therefore
 69 perform the MCMC evolution in mirror space [41], in which Langevin dynamics evaluated through a
 70 suitable mirror map retain the convergence properties of the corresponding unconstrained algorithm.
 71 We use the element-wise transform $y_i = \Phi^{-1}(x_i) = \sqrt{2} \operatorname{erf}^{-1}(2x_i - 1)$. The mirrored target density
 72 is the corresponding change of variables

$$\log \tilde{p}_{\text{mir}}(y) = \log \tilde{p}(\Phi(y)) - \frac{1}{2} \|y\|^2 + \text{const.} \quad (3)$$

73 Note that the constant drops out in the score.

¹Code is provided in the anonymized supplementary material.

74 2.3 Learned Stein Discrepancy

75 To monitor convergence of the ULA chains, we need a distributional discrepancy that can be evaluated
 76 using samples from the chain and the unnormalized density of the target. The Stein discrepancy
 77 provides such a diagnostic [42, 43]. For a vector-valued critic function $f : \mathbb{R}^d \rightarrow \mathbb{R}^d$, the Stein
 78 discrepancy between a smooth, sampled distribution p and a smooth target distribution q is

$$S(p, q) = \sup_{f \in \mathcal{F}} \mathbb{E}_{x \sim p} [s_q(x)^T f(x) + \nabla \cdot f(x)], \quad (4)$$

79 where $\mathcal{F}$ is a norm-bounded class of differentiable functions for which the expectation over p is finite
 80 and $q(x)f(x)$ has vanishing tails. For sufficiently rich critic classes, this quantity is non-negative and
 81 satisfies $S(p, q) = 0$ if and only if $p = q$ [44].

82 Kernel Stein discrepancies are a popular choice for $\mathcal{F}$ [45–47], but their standard estimators involve
 83 pairwise sums that scale as n^2 in the number of samples n . Since we typically work in a regime with
 84 large numbers of samples, n^2 evaluation is infeasible, and a more natural choice would be one that
 85 exploits this large data size. We thus use the learned Stein discrepancy (LSD) [48, 49], taking $\mathcal{F}$ to
 86 be a class of square-integrable differentiable functions that satisfy the Stein boundary condition, and
 87 representing the critic by a neural network f_ϕ . In practice, square integrability is enforced by adding
 88 a quadratic regularization term to the training objective,

$$\mathcal{L}_\lambda(\phi; p, q) = -\mathbb{E}_{x \sim p} [s_q(x)^T f_\phi(x) + \nabla \cdot f_\phi(x) - \lambda \|f_\phi(x)\|^2]. \quad (5)$$

89 After training, the learned Stein discrepancy is evaluated with the learned critic on a held-out sample,
 90 without the regularization term.

91 2.4 Sharp Boundaries

92 Physical cross sections typically involve sharp phase-space cuts, which restrict the target density to
 93 an allowed region with a hard boundary. The derivation of Eq. 4 involves an integration by parts over
 94 this region, and a boundary contribution will thus remain unless the Stein test function has vanishing
 95 flux through the cut surface [50, 51]. We address this issue by introducing a scalar boundary function
 96 $h(x)$ that is positive inside the allowed region and vanishes on the cut boundary. The Stein operator
 97 is then applied to the vector field hf rather than to f itself [50, 52],

$$S_h(p, q) = \sup_{f \in \mathcal{F}} \mathbb{E}_{x \sim p} [h(x)s_q(x)^T f(x) + h(x)\nabla \cdot f(x) + \nabla h(x)^T f(x)]. \quad (6)$$

98 2.5 Neural Surrogates

99 The sampling strategy described above relies on gradients of the target density. Differentiable matrix
 100 element implementations have recently become available [29, 53–56], enabling the sampling strategy
 101 set out above, but at higher computational expense. This motivates us to consider the use of learned
 102 surrogates to reduce the required number of evaluations.

103 Neural surrogates for scattering amplitudes and matrix elements have been studied extensively [34–
 104 36, 57–71]. In the present setting, one can train a surrogate on a relatively limited set of points without
 105 requiring gradients. Then, chains can be evolved cheaply through the early part of the trajectory using
 106 the surrogate as target, before switching over to a shorter sequence of ULA steps using the exact
 107 target. The surrogate is thus only used to accelerate equilibration, while the final evolution remains
 108 tied to the physical target density. We investigate this strategy in Section 3.

109 3 Results

110 We validate the learned Stein diagnostic on analytically tractable Ornstein–Uhlenbeck targets in
 111 Appendix B.

112 We apply ULA and the LSD diagnostic to sample the cross section for the process $u\bar{u} \rightarrow Z + ng$.
 113 Matrix elements and their gradients are evaluated using MadJax [53] with several modifications for
 114 numerical stability. We use the NNPDF4.0 LO set [72], and implement the PDF and α_s interpolation
 115 in JAX [73] following the LHAPDF interpolation prescription [74]. We set $\sqrt{s} = 13$ TeV and apply
 116 cuts $p_T > 20$ GeV and $|y| < 5$ to all final-state particles and a separation cut $\Delta R > 0.4$ to the

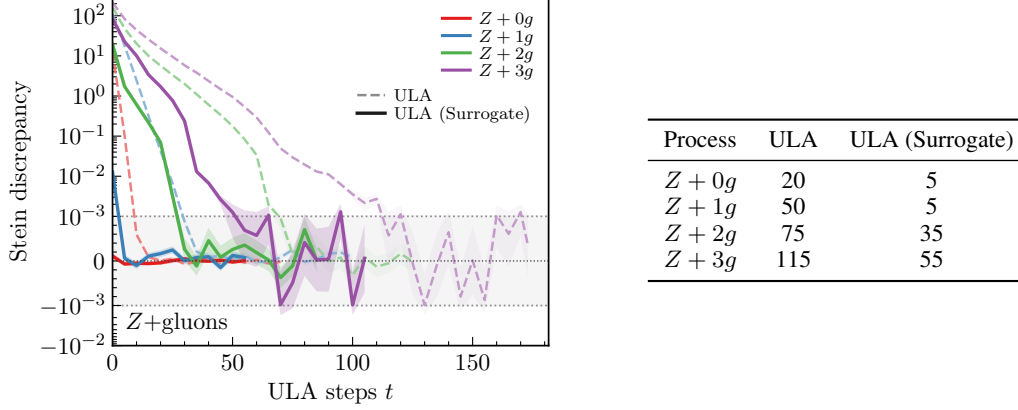


Figure 1: Convergence and relaxation times. Left: the LSD as a function of the number of ULA steps for $Z + ng$ cross sections. Colours denote gluon multiplicities, while line styles distinguish ULA with and without surrogate initialization. Right: relaxation-time checkpoints t_{rel} .

117 gluons. Phase space is mapped to the $[0, 1]^{3n+1}$ hypercube using the Chili [75] vector boson plus jets
 118 map. The Stein diagnostic uses the same critic architecture and optimization strategy as in the OU
 119 validation, with $\lambda = 0.1$. The boundary function is

$$h(x) = \left[\frac{1}{2 + \binom{n}{2}} \left(\frac{1}{1 - x_a} + \frac{1}{1 - x_b} + \sum_{1 \leq i < j \leq n} \frac{1}{1 - \Delta R_{\text{cut}}^2 / \Delta R_{ij}^2} \right) \right]^{-1}, \quad (7)$$

120 for points that pass the cuts, and 0 for points that do not.

121 We use ULA and the LSD diagnostic to generate 10^7 samples for $n = 0, 1, 2, 3$, both with and
 122 without neural surrogates. The surrogate is a residual MLP with five hidden layers of width 128,
 123 GeLU activations, and a scalar output. It is trained to predict the log cross section from only 10^6
 124 samples using Adam and a mean-squared error loss. We then run ULA on the surrogate for a large
 125 number of steps. The terminal states are then used as the initial states for the ULA run, which
 126 is otherwise initialized with unit Gaussian samples. We define the relaxation time t_{rel} as the first
 127 checkpoint for which the learned LSD is no more than one standard error above zero, $S_{\text{LSD}} \leq \Delta S_{\text{LSD}}$.
 128 After the trigger, we evolve the chain for a further $\max(0.5 t_{\text{rel}}, 50)$ ULA steps. Figure 1 shows
 129 the convergence results and summarizes the relaxation-time triggers. The surrogate warm start
 130 substantially reduces the number of exact-target ULA steps required for relaxation. All reported
 131 experiments were run on a single NVIDIA GeForce RTX 5090 GPU (32 GB) and required less than
 132 24 hours in total.

133 Figure 2 compares ULA samples with and without surrogate initialization against independent VEGAS
 134 estimates for four physical observables. The jet-resolution scale $\sqrt{d_{01}}$ is obtained using exclusive
 135 k_t clustering with $R = 1$ [76, 77]. In every case, the terminal ULA samples reproduce the VEGAS
 136 distributions within statistical uncertainties.

137 4 Discussion

138 The results above suggest that gradient-based parallel MCMC is a practically viable alternative to
 139 rejection-based event generation, at least in the regime studied here. The central practical quantity is
 140 the number of exact matrix-element evaluations needed before terminal samples can be used. For
 141 the $Z + ng$ examples, the learned Stein diagnostic triggers after a modest number of exact-target
 142 ULA steps, and the increase with multiplicity is mild over the range considered. This is encouraging
 143 because unweighting efficiencies can degrade by orders of magnitude as the final-state multiplicity
 144 increases [34, 35]. Furthermore, the methods explored here generate strictly unweighted events, while
 145 in the usual rejection sampling setup residual weights may remain when using the maximum weight
 146 would otherwise reduce the efficiency even further. Like conventional rejection-based generation, our
 147 method is also naturally parallel: each chain can be evolved independently, while the adaptation and

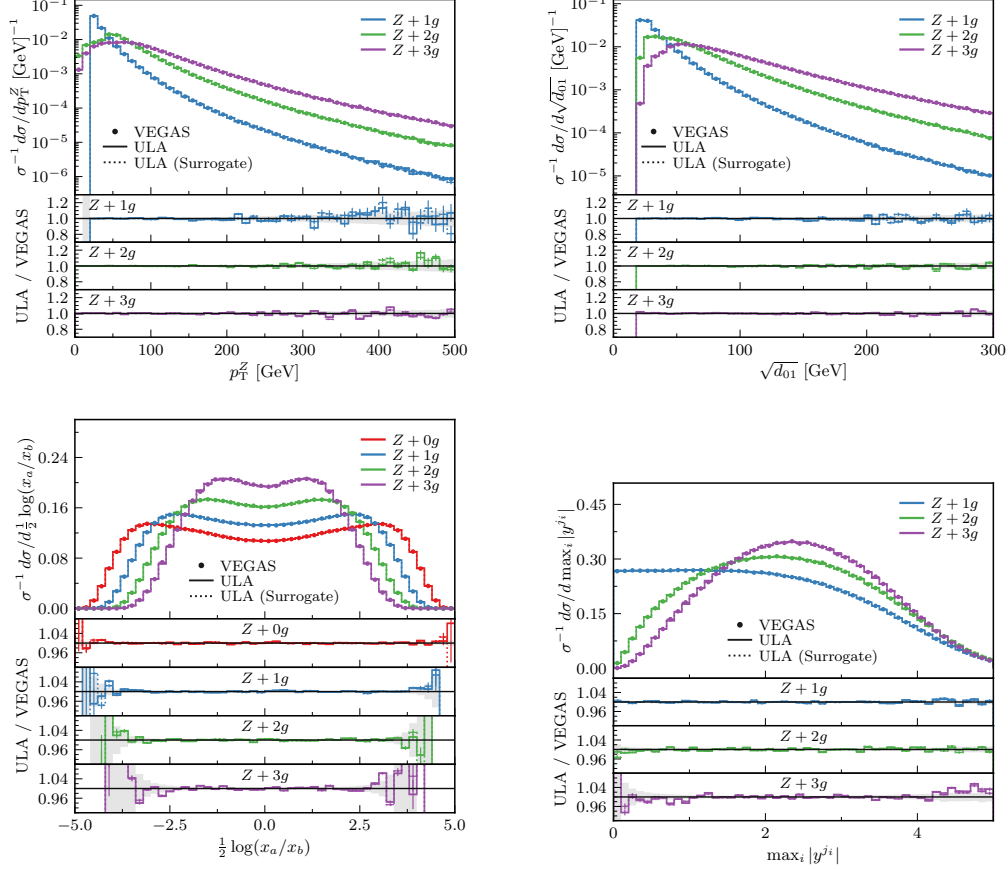


Figure 2: Normalized differential distributions for the Z -boson transverse momentum, the exclusive k_t clustering scale $\sqrt{d_{01}}$, the full-system boost rapidity $\frac{1}{2} \log(x_a/x_b)$, and the largest absolute gluon rapidity $\max_i |y^{j_i}|$. Filled markers show the VEGAS reference, solid lines show ULA samples, and dotted lines show ULA samples after surrogate initialization. The lower panels show the corresponding ratios to VEGAS for each multiplicity.

Stein diagnostic require only ensemble-level reductions. This makes the method well-matched to GPU/TPU execution.

The required number of evaluations can be further reduced through the use of surrogate initialization. In our implementation the surrogate is deliberately simple: it uses the same network architecture as the Stein critic and is trained on a comparatively small set of uniformly sampled phase-space points. Nevertheless, it substantially shortens the exact-target relaxation time. This should be viewed as a conservative demonstration rather than an optimized surrogate strategy. More expressive architectures, improved training samples, or active refinement in the regions explored by the chains could plausibly make the warm start closer to the physical target and further reduce the number of exact matrix-element and gradient evaluations.

Reverse-mode automatic differentiation requires one forward and backward pass, so for tree-level matrix elements composed of rational functions, gradient evaluation should approach a small constant-factor overhead over regular evaluation [78, 79].

The present study considers a fixed partonic channel, tree-level matrix elements, and a simple phase-space map. The observed scaling, and more generally any comparison with current methods, can only be tested in the context of an implementation in full event generator programs. Still, the combination of parallel chains, an explicit relaxation-time diagnostic, and surrogate initialization provides a concrete route towards event generation in which the expensive exact target is evaluated as few times as possible.

References

- [1] I. Zurbano Fernandez et al., *High-Luminosity Large Hadron Collider (HL-LHC): Technical design report*, no. 10/2020 in CERN Yellow Reports: Monographs, CERN, Geneva (12, 2020), [10.23731/CYRM-2020-0010](#).
- [2] FCC collaboration, *FCC-ee: The Lepton Collider: Future Circular Collider Conceptual Design Report Volume 2*, *Eur. Phys. J. ST* **228** (2019) 261.
- [3] FCC collaboration, *FCC-hh: The Hadron Collider: Future Circular Collider Conceptual Design Report Volume 3*, *Eur. Phys. J. ST* **228** (2019) 755.
- [4] A. Buckley et al., *General-purpose event generators for LHC physics*, *Phys. Rept.* **504** (2011) 145 [[1101.2599](#)].
- [5] M. Narain et al., *The Future of US Particle Physics - The Snowmass 2021 Energy Frontier Report*, [2211.11084](#).
- [6] HEP SOFTWARE FOUNDATION collaboration, *A Roadmap for HEP Software and Computing R&D for the 2020s*, *Comput. Softw. Big Sci.* **3** (2019) 7 [[1712.06982](#)].
- [7] HSF PHYSICS EVENT GENERATOR WG collaboration, *Challenges in Monte Carlo Event Generator Software for High-Luminosity LHC*, *Comput. Softw. Big Sci.* **5** (2021) 12 [[2004.13687](#)].
- [8] European Strategy Group, *2020 Update of the European Strategy for Particle Physics*, CERN Council, Geneva (2020), [10.17181/ESU2020](#).
- [9] M. van Beekveld, E. Bothmann, A. Buckley, C. Gütschow, P. Skands and R. Winterhalder, *The Monte Carlo Ecosystem in High-Energy Physics: A Primer*, [2605.16036](#).
- [10] G.P. Lepage, *A New Algorithm for Adaptive Multidimensional Integration*, *J. Comput. Phys.* **27** (1978) 192.
- [11] G.P. Lepage, *Adaptive multidimensional integration: VEGAS enhanced*, *J. Comput. Phys.* **439** (2021) 110386 [[2009.05112](#)].
- [12] R. Kleiss and R. Pittau, *Weight optimization in multichannel Monte Carlo*, *Comput. Phys. Commun.* **83** (1994) 141 [[hep-ph/9405257](#)].
- [13] T. Ohl, *Vegas revisited: Adaptive Monte Carlo integration beyond factorization*, *Comput. Phys. Commun.* **120** (1999) 13 [[hep-ph/9806432](#)].
- [14] F. Maltoni and T. Stelzer, *MadEvent: Automatic event generation with MadGraph*, *JHEP* **02** (2003) 027 [[hep-ph/0208156](#)].
- [15] F. Krauss, R. Kuhn and G. Soff, *AMEGIC++ 1.0: A Matrix element generator in C++*, *JHEP* **02** (2002) 044 [[hep-ph/0109036](#)].
- [16] C.G. Papadopoulos, *PHEGAS: A Phase space generator for automatic cross-section computation*, *Comput. Phys. Commun.* **137** (2001) 247 [[hep-ph/0007335](#)].
- [17] W. Kilian, T. Ohl and J. Reuter, *WHIZARD: Simulating Multi-Particle Processes at LHC and ILC*, *Eur. Phys. J. C* **71** (2011) 1742 [[0708.4233](#)].
- [18] S. Jadach, *Foam: A General purpose cellular Monte Carlo event generator*, *Comput. Phys. Commun.* **152** (2003) 55 [[physics/0203033](#)].
- [19] T. Gleisberg and S. Hoeche, *Comix, a new matrix element generator*, *JHEP* **12** (2008) 039 [[0808.3674](#)].
- [20] SHERPA collaboration, *Event generation with Sherpa 3*, *JHEP* **12** (2024) 156 [[2410.22148](#)].
- [21] S. Höche, S. Prestel and H. Schulz, *Simulation of Vector Boson Plus Many Jet Final States at the High Luminosity LHC*, *Phys. Rev. D* **100** (2019) 014024 [[1905.05120](#)].

- [22] M.D. Klimek and M. Perelstein, *Neural Network-Based Approach to Phase Space Integration*, *SciPost Phys.* **9** (2020) 053 [[1810.11509](#)].
- [23] E. Bothmann, T. Janßen, M. Knobbe, T. Schmale and S. Schumann, *Exploring phase space with Neural Importance Sampling*, *SciPost Phys.* **8** (2020) 069 [[2001.05478](#)].
- [24] C. Gao, S. Höche, J. Isaacson, C. Krause and H. Schulz, *Event Generation with Normalizing Flows*, *Phys. Rev. D* **101** (2020) 076002 [[2001.10028](#)].
- [25] B. Stienen and R. Verheyen, *Phase space sampling and inference from weighted events with autoregressive flows*, *SciPost Phys.* **10** (2021) 038 [[2011.13445](#)].
- [26] T. Heime1, R. Winterhalder, A. Butter, J. Isaacson, C. Krause, F. Maltoni et al., *MadNIS - Neural multi-channel importance sampling*, *SciPost Phys.* **15** (2023) 141 [[2212.06172](#)].
- [27] R. Verheyen, *Event Generation and Density Estimation with Surjective Normalizing Flows*, *SciPost Phys.* **13** (2022) 047 [[2205.01697](#)].
- [28] T. Heime1, N. Huetsch, F. Maltoni, O. Mattelaer, T. Plehn and R. Winterhalder, *The MadNIS Reloaded*, *arXiv preprint* (2023) [[2311.01548](#)].
- [29] T. Heime1, O. Mattelaer, T. Plehn and R. Winterhalder, *Differentiable MadNIS-Lite*, *SciPost Phys.* **18** (2025) 017 [[2408.01486](#)].
- [30] H. Kharraziha and S. Moretti, *The Metropolis algorithm for on-shell four momentum phase space*, *Comput. Phys. Commun.* **127** (2000) 242 [[hep-ph/9909313](#)].
- [31] S. Weinzierl, *A General algorithm to generate unweighted events for next-to-leading order calculations in electron positron annihilation*, *JHEP* **08** (2001) 028 [[hep-ph/0106146](#)].
- [32] K. Kroeninger, S. Schumann and B. Willenberg, *(MC)**3 – a Multi-Channel Markov Chain Monte Carlo algorithm for phase-space sampling*, *Comput. Phys. Commun.* **186** (2015) 1 [[1404.4328](#)].
- [33] S. La Cagnina, C. Grunwald, T. Janßen, K. Kröninger and S. Schumann, *Phase space sampling with Markov Chain Monte Carlo methods*, *EPJ Web Conf.* **337** (2025) 01319 [[2412.12963](#)].
- [34] K. Danziger, T. Janßen, S. Schumann and F. Siegert, *Accelerating Monte Carlo event generation – rejection sampling using neural network event-weight estimates*, *SciPost Phys.* **12** (2022) 164 [[2109.11964](#)].
- [35] T. Janßen, D. Maître, S. Schumann, F. Siegert and H. Truong, *Unweighting multijet event generation using factorisation-aware neural networks*, *SciPost Phys.* **15** (2023) 107 [[2301.13562](#)].
- [36] T. Herrmann, T. Janßen, M. Schenker, S. Schumann and F. Siegert, *Accelerating multijet-merged event generation with neural network matrix element surrogates*, *SciPost Phys.* **20** (2026) 071 [[2506.06203](#)].
- [37] R.M. Neal, *MCMC using Hamiltonian dynamics*, in *Handbook of Markov Chain Monte Carlo*, pp. 113–162, Chapman and Hall/CRC (2011) [[1206.1901](#)].
- [38] M.D. Hoffman and A. Gelman, *The No-U-Turn Sampler: Adaptively setting path lengths in Hamiltonian Monte Carlo*, *J. Mach. Learn. Res.* **15** (2014) 1593 [[1111.4246](#)].
- [39] G.O. Roberts and R.L. Tweedie, *Exponential convergence of Langevin distributions and their discrete approximations*, *Bernoulli* **2** (1996) 341.
- [40] X. Cheng, N.S. Chatterji, P.L. Bartlett and M.I. Jordan, *Underdamped Langevin MCMC: A non-asymptotic analysis*, in *Conference on Learning Theory*, pp. 300–323, PMLR, 2018 [[1707.03663](#)].
- [41] Y.-P. Hsieh, A. Kavis, P. Rolland and V. Cevher, *Mirrored Langevin Dynamics*, in *Advances in Neural Information Processing Systems*, vol. 31, 2018 [[1802.10174](#)].

- [42] J. Gorham and L. Mackey, *Measuring Sample Quality with Stein’s Method*, in *Advances in Neural Information Processing Systems*, vol. 28, 2015 [[1506.03039](#)].
- [43] A. Anastasiou, A. Barp, F.-X. Briol, B. Ebner, R.E. Gaunt, F. Ghaderinezhad et al., *Stein’s Method Meets Computational Statistics: A Review of Some Recent Developments*, *Statistical Science* **38** (2023) 120 [[2105.03481](#)].
- [44] C. Stein, P. Diaconis, S. Holmes and G. Reinert, *Use of Exchangeable Pairs in the Analysis of Simulations*, *Lecture Notes-Monograph Series* (2004) 1.
- [45] Q. Liu, J. Lee and M. Jordan, *A Kernelized Stein Discrepancy for Goodness-of-Fit Tests*, in *International Conference on Machine Learning*, pp. 276–284, PMLR, 2016 [[1602.03253](#)].
- [46] K. Chwialkowski, H. Strathmann and A. Gretton, *A Kernel Test of Goodness of Fit*, in *International Conference on Machine Learning*, pp. 2606–2615, PMLR, 2016 [[1602.02964](#)].
- [47] J. Gorham and L. Mackey, *Measuring Sample Quality with Kernels*, in *International Conference on Machine Learning*, pp. 1292–1301, PMLR, 2017 [[1703.01717](#)].
- [48] T. Hu, Z. Chen, H. Sun, J. Bai, M. Ye and G. Cheng, *Stein Neural Sampler*, [1810.03545](#).
- [49] W. Grathwohl, K.-C. Wang, J.-H. Jacobsen, D. Duvenaud and R. Zemel, *Learning the Stein Discrepancy for Training and Evaluating Energy-Based Models Without Sampling*, in *International Conference on Machine Learning*, pp. 3732–3747, PMLR, 2020 [[2002.05616](#)].
- [50] S. Liu, T. Kanamori and D.J. Williams, *Estimating Density Models with Truncation Boundaries Using Score Matching*, *Journal of Machine Learning Research* **23** (2022) 1 [[1910.03834](#)].
- [51] D.J. Williams and S. Liu, *Approximate Stein Classes for Truncated Density Estimation*, in *International Conference on Machine Learning*, pp. 37066–37090, PMLR, 2023 [[2306.00602](#)].
- [52] W. Xu, *Standardisation-Function Kernel Stein Discrepancy: A Unifying View on Kernel Stein Discrepancy Tests for Goodness-of-Fit*, in *International Conference on Artificial Intelligence and Statistics*, pp. 1575–1597, PMLR, 2022.
- [53] L. Heinrich and M. Kagan, *Differentiable Matrix Elements with MadJax*, *J. Phys. Conf. Ser.* **2438** (2023) 012137 [[2203.00057](#)].
- [54] S. Carrazza, J. Cruz-Martinez, M. Rossi and M. Zaro, *MadFlow: towards the automation of Monte Carlo simulation on GPU for particle physics processes*, *EPJ Web Conf.* **251** (2021) 03022 [[2105.10529](#)].
- [55] S. Carrazza, J. Cruz-Martinez, M. Rossi and M. Zaro, *MadFlow: automating Monte Carlo simulation on GPU for particle physics processes*, *Eur. Phys. J. C* **81** (2021) 656 [[2106.10279](#)].
- [56] T. Heimel, O. Mattelaer and R. Winterhalder, *MadSpace – Event Generation for the Era of GPUs and ML*, [2602.06895](#).
- [57] F. Bishara and M. Montull, *Machine learning amplitudes for faster event generation*, *Phys. Rev. D* **107** (2023) L071901 [[1912.11055](#)].
- [58] S. Badger and J. Bullock, *Using neural networks for efficient evaluation of high multiplicity scattering amplitudes*, *JHEP* **06** (2020) 114 [[2002.07516](#)].
- [59] J. Aylett-Bullock, S. Badger and R. Moodie, *Optimising simulations for diphoton production at hadron colliders using amplitude neural networks*, *JHEP* **08** (2021) 066 [[2106.09474](#)].
- [60] D. Maître and H. Truong, *A factorisation-aware Matrix element emulator*, *JHEP* **11** (2021) 066 [[2107.06625](#)].
- [61] S. Badger, A. Butter, M. Luchmann, S. Pitz and T. Plehn, *Loop amplitudes from precision networks*, *SciPost Phys. Core* **6** (2023) 034 [[2206.14831](#)].

- [62] D. Maître and H. Truong, *One-loop matrix element emulation with factorisation awareness*, *JHEP* **05** (2023) 159 [2302.04005].
- [63] H. Bahl, N. Elmer, L. Favaro, M. Haussmann, T. Plehn and R. Winterhalder, *Accurate surrogate amplitudes with calibrated uncertainties*, *SciPost Phys. Core* **8** (2025) 073 [2412.12069].
- [64] J. Brehmer, V. Bresó, P. de Haan, T. Plehn, H. Qu, J. Spinner et al., *A Lorentz-equivariant transformer for all of the LHC*, *SciPost Phys.* **19** (2025) 108 [2411.00446].
- [65] V. Bresó-Pla, G. Heinrich, V. Magerya and A. Olsson, *Interpolating amplitudes*, *SciPost Phys.* **19** (2025) 123 [2412.09534].
- [66] H. Bahl, N. Elmer, T. Plehn and R. Winterhalder, *Amplitude Uncertainties Everywhere All at Once*, *SciPost Phys.* **20** (2026) 083 [2509.00155].
- [67] L. Favaro, G. Gerhartz, F.A. Hamprecht, P. Lippmann, S. Pitz, T. Plehn et al., *Lorentz-Equivariance without Limitations*, 2508.14898.
- [68] J.M. Villadamigo, R. Frederix, T. Plehn, T. Vitos and R. Winterhalder, *FASTColor – Full-color Amplitude Surrogate Toolkit for QCD*, 2509.07068.
- [69] L. Beccatini, F. Maltoni, O. Mattelaer and R. Winterhalder, *Amplitude Surrogates for Multi-Jet Processes*, 2512.11036.
- [70] H. Bahl, V. Bresó-Pla, A. Butter and J.I. Ramirez, *Scaling laws for amplitude surrogates*, 2601.13308.
- [71] H. Bahl, J. Braun, G. Heinrich, T. Plehn and R. Revelli, *How to Trust Learned Loop Amplitudes*, 2601.00950.
- [72] NNPDF collaboration, *The path to proton structure at 1% accuracy*, *Eur. Phys. J. C* **82** (2022) 428 [2109.02653].
- [73] J. Bradbury, R. Frostig, P. Hawkins, M.J. Johnson, Y. Katariya, C. Leary et al., “JAX: composable transformations of Python+NumPy programs.” <https://github.com/google/jax>, 2018.
- [74] A. Buckley, J. Ferrando, S. Lloyd, K. Nordström, B. Page, M. Rüfenacht et al., *LHAPDF6: parton density access in the LHC precision era*, *Eur. Phys. J. C* **75** (2015) 132 [1412.7420].
- [75] E. Bothmann, T. Childers, W. Giele, F. Herren, S. Hoeche, J. Isaacson et al., *Efficient phase-space generation for hadron collider event simulation*, *SciPost Phys.* **15** (2023) 169 [2302.10449].
- [76] S. Catani, Y.L. Dokshitzer, M.H. Seymour and B.R. Webber, *Longitudinally invariant K_t clustering algorithms for hadron hadron collisions*, *Nucl. Phys. B* **406** (1993) 187.
- [77] S.D. Ellis and D.E. Soper, *Successive combination jet algorithm for hadron collisions*, *Phys. Rev. D* **48** (1993) 3160 [hep-ph/9305266].
- [78] W. Baur and V. Strassen, *The complexity of partial derivatives*, *Theoretical Computer Science* **22** (1983) 317.
- [79] A. Griewank, *Who invented the reverse mode of differentiation?*, in *Optimization stories*, pp. 389–400, European Mathematical Society Publishing House (2012).
- [80] N. Brosse, A. Durmus, É. Moulines and S. Sabanis, *The Tamed Unadjusted Langevin Algorithm*, *Stochastic Processes and their Applications* **129** (2019) 3638 [1710.05559].
- [81] M. Girolami and B. Calderhead, *Riemann Manifold Langevin and Hamiltonian Monte Carlo Methods*, *Journal of the Royal Statistical Society Series B: Statistical Methodology* **73** (2011) 123.
- [82] M.K. Titsias, *Optimal Preconditioning and Fisher Adaptive Langevin Sampling*, in *Advances in Neural Information Processing Systems*, vol. 36, 2023 [2305.14442].

- 346 [83] A. Gelman, W.R. Gilks and G.O. Roberts, *Weak Convergence and Optimal Scaling of Random*
347 *Walk Metropolis Algorithms*, *The Annals of Applied Probability* **7** (1997) 110.
- 348 [84] C. Pasarica and A. Gelman, *Adaptively scaling the metropolis algorithm using expected*
349 *squared jumped distance*, *Statistica Sinica* (2010) 343.
- 350 [85] H. Robbins and S. Monro, *A Stochastic Approximation Method*, *The Annals of Mathematical*
351 *Statistics* (1951) 400.
- 352 [86] G.O. Roberts and J.S. Rosenthal, *Coupling and ergodicity of adaptive Markov chain Monte*
353 *Carlo algorithms*, *Journal of Applied Probability* **44** (2007) 458.
- 354 [87] D.P. Kingma and J. Ba, *Adam: A Method for Stochastic Optimization*, in *International*
355 *Conference on Learning Representations*, 2015 [[1412.6980](#)].
- 356 [88] S. Chewi, C. Lu, K. Ahn, X. Cheng, T. Le Gouic and P. Rigollet, *Optimal Dimension*
357 *Dependence of the Metropolis-Adjusted Langevin Algorithm*, in *Conference on Learning*
358 *Theory*, pp. 1260–1300, PMLR, 2021 [[2012.12810](#)].

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [\[Yes\]](#)

Justification: The abstract and Introduction state the algorithmic contribution, experimental setting, and principal empirical findings. The Discussion limits the scope to a fixed partonic channel, tree-level matrix elements, and a simple phase-space map.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [\[Yes\]](#)

Justification: The Discussion identifies the restricted process, perturbative order, and phase-space map, and states that scaling and comparisons with production methods require implementation in a full event generator.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [\[N/A\]](#)

Justification: The paper does not claim a new theorem or formal theoretical guarantee. Standard results used to construct the sampler and Stein diagnostic are cited, while the analytically tractable validation is described in Appendix B.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [\[Yes\]](#)

Justification: The Results and Appendices A–B specify the physical process, cuts, sample counts, sampler updates, diagnostic, surrogate setup, and validation. The anonymized supplementary archive provides the implementation, trained models, numerical results, and plotting scripts.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [\[Yes\]](#)

Justification: The anonymized supplementary archive provides the source code, trained surrogate models, numerical results, plotting scripts, environment specification, and exact commands for regenerating every reported experiment and figure.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [\[Yes\]](#)

Justification: The Results and Appendix B give the physics setup, cuts, sample counts, regularization, critic architecture, optimizer, and validation procedure. Appendix A and the supplementary code provide the remaining sampler and implementation details.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [\[Yes\]](#)

Justification: The figures report one-standard-error uncertainty bars for the Monte Carlo estimates, and the relaxation criterion is explicitly defined in terms of one standard error.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: The Results report the GPU model and memory and state that the complete set of reported experiments required less than 24 hours in total.

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: The work uses public scientific software and simulated physics data, involves no human subjects or sensitive personal data, and is submitted with anonymized supplementary material.

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [N/A]

Justification: This is foundational work on particle-physics simulation with no direct deployment or identified path to material positive or negative societal impacts beyond general scientific and computational benefits.

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The released code, numerical results, and task-specific surrogate models do not constitute high-risk or dual-use assets requiring release safeguards.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: The paper cites the scientific sources, while the anonymized supplementary archive identifies the provenance, version, authors, source, license or published terms, modifications, and required notices for bundled third-party assets and direct software dependencies.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The anonymized archive documents the new code, trained models, numerical results, training setup, limitations, and complete reproduction procedure, and releases the original code under the MIT License.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

458 Answer: [N/A]
459 Justification: The research does not involve crowdsourcing or human subjects.
460 15. **Institutional review board (IRB) approvals or equivalent for research with human**
461 **subjects**
462 Question: Does the paper describe potential risks incurred by study participants, whether
463 such risks were disclosed to the subjects, and whether Institutional Review Board (IRB)
464 approvals (or an equivalent approval/review based on the requirements of your country or
465 institution) were obtained?
466 Answer: [N/A]
467 Justification: The research does not involve crowdsourcing or human subjects, so IRB or
468 equivalent approval is not applicable.
469 16. **Declaration of LLM usage**
470 Question: Does the paper describe the usage of LLMs if it is an important, original, or
471 non-standard component of the core methods in this research? Note that if the LLM is used
472 only for writing, editing, or formatting purposes and does *not* impact the core methodology,
473 scientific rigor, or originality of the research, declaration is not required.
474 Answer: [N/A]
475 Justification: LLMs are not an important, original, or non-standard component of the core
476 methods or experiments in this work.

477 A Underdamped Langevin algorithm

478 In this appendix, we describe the Underdamped Langevin algorithm implementation in detail. Each
 479 MCMC chain carries a position x and a velocity v . We assume we have access to the score
 480 $s(x) = \nabla \log \tilde{p}(x)$. For stability purposes the score is regularized component-wise to $\tilde{s}_i(x) =$
 481 $s_i(x)/(1 + \eta_i |s_i(x)|)$, where η_i are the MCMC step sizes [80]. At time step t , a new proposal x', v'
 482 is generated following the leapfrog step

$$\begin{aligned} v_{1/2} &= v_t + \frac{1}{2} \Lambda_\eta \tilde{s}(x_t), \\ x' &= x_t + \Lambda_\eta v_{1/2}, \\ v' &= v_{1/2} + \frac{1}{2} \Lambda_\eta \tilde{s}(x'), \end{aligned} \quad (8)$$

483 where $\Lambda_\eta = \text{diag}(\eta_1, \dots, \eta_d)$. This step discretizes the Hamiltonian dynamics on the augmented
 484 density

$$\pi(x, v) \propto \tilde{p}(x) \exp\left(-\frac{1}{2} v^T v\right). \quad (9)$$

485 In the continuous-time limit, the Hamiltonian flow conserves the Hamiltonian $H(x, v) = -\log \tilde{p}(x) +$
 486 $v^T v/2$ and preserves phase-space volume. As a result, $\pi(x, v)$ as well as its x -marginal, which is
 487 proportional to the desired target density $\tilde{p}(x)$, is invariant. However, at finite step sizes an integration
 488 error is introduced. By applying a MH acceptance step with probability

$$a = \min\{1, \exp[-H(x', v') + H(x_t, v_t)]\} \quad (10)$$

489 the discretization bias is corrected. Upon rejection, the position is kept and the incoming momentum
 490 is reversed $v_t \rightarrow -v_t$. After the MH step, the momentum is refreshed

$$v_{t+1} = \beta v_{\text{acc}} + \sqrt{1 - \beta^2} \zeta, \quad \zeta \sim \mathcal{N}(0, I). \quad (11)$$

491 These momentum updates preserve the Gaussian velocity marginal and inject randomness into the
 492 otherwise deterministic Hamiltonian evolution, allowing the chain to move between trajectories and
 493 explore the full target distribution. After equilibration, terminal positions drawn from these chains
 494 are distributed according to $\tilde{p}(x)$.

495 We write the step sizes as $\log \eta_i = \log \epsilon + \log \rho_i$, with $\sum_i \log \rho_i = 0$. The scalar scale ϵ is
 496 initialized by doubling or halving until the average acceptance probability is close to one half [38],
 497 while the directions ρ_i are initialized from the diagonal score second moment, motivated by Fisher
 498 preconditioning [81, 82],

$$F_{ii,t} = \mathbb{E}_{x_t} [s_i(x_t)^2], \quad \log \rho_i = -\frac{1}{2} \log F_{ii,t} + \frac{1}{2d} \sum_{j=1}^d \log F_{jj,t}. \quad (12)$$

499 This fixes $\prod_i \rho_i = 1$ and increases step sizes in directions with smaller typical score variation.
 500 During the chain, $\log \epsilon$ is adapted by ascending the expected squared jump distance $\text{ESJD}(\epsilon) =$
 501 $\mathbb{E}_{x_t, v_t, \zeta} [a \|x' - x_t\|^2]$ at fixed ρ , a standard proxy for MCMC efficiency [83, 84]. The directions are
 502 adapted by updating ρ_i with new estimates of the diagonal score second moment. Both the ESJD scale
 503 update and the score-moment direction update use the same diminishing Robbins–Monro schedule,
 504 so that the transition-kernel changes vanish asymptotically, as required for stationarity [85, 86]. All
 505 expectations are evaluated over the parallel chain ensemble.

506 B Stein diagnostic validation

507 In this appendix, we validate the Stein diagnostic on a target for which the MCMC convergence can
 508 be computed analytically. Starting from a fixed point μ_0 away from some target Gaussian mean μ_q
 509 and diagonal target Σ_q , we sample an exact Ornstein–Uhlenbeck (OU) transition kernel

$$x_{t+1} = \mu_q + e^{-\eta \Sigma_q^{-1}} (x_t - \mu_q) + \Sigma_q^{1/2} \left(I - e^{-2\eta \Sigma_q^{-1}}\right)^{1/2} \xi, \quad \xi \sim \mathcal{N}(0, I), \quad (13)$$

where η is the step size. After t steps, the marginal distribution is Gaussian with

$$\begin{aligned}\mu_t &= \mu_q + e^{-t\eta\Sigma_q^{-1}}(\mu_0 - \mu_q) \\ \Sigma_t &= \Sigma_q \left(I - e^{-2t\eta\Sigma_q^{-1}} \right).\end{aligned}\tag{14}$$

Thus p_t converges to the target density as $t \rightarrow \infty$, yielding a controlled and testable sequence of distributions. To validate the role of the boundary function, we apply a radial cut that removes 20% of the target probability mass centered around the target mean μ_q .

Under Eq. (5), the optimal Stein discrepancy is [48]

$$S_{\text{opt}}(p_t, q) = \frac{1}{2\lambda} \mathbb{E}_{x \sim p_t} \left[\|s_q(x) - s_{p_t}(x)\|^2 \right].\tag{15}$$

We compare this optimal value against the LSD computed on OU samples and on ULA samples. The validation results use dimensions $d = 4$ and $d = 16$, with target parameters

$$\mu_q = \text{linspace}(-2, 2, d), \quad \Sigma_q = \text{diag}(\text{linspace}(0.5, 2, d)^2).\tag{16}$$

The OU process is initialized from the fixed point $\mu_0 = \mu_q + 3$, while ULA is initialized from a unit Gaussian. The LSD regularization is $\lambda = 0.1$. The Stein critic is modelled with a five-layer residual GeLU network with hidden width 128. Optimization is performed with the Adam optimizer [87] on $n = 10^7$ chains, and final LSD values are obtained using 10-fold cross validation. The boundary function is $h(x) = \max\{1 - r_{\text{cut}}^2/r^2(x), 0\}$.

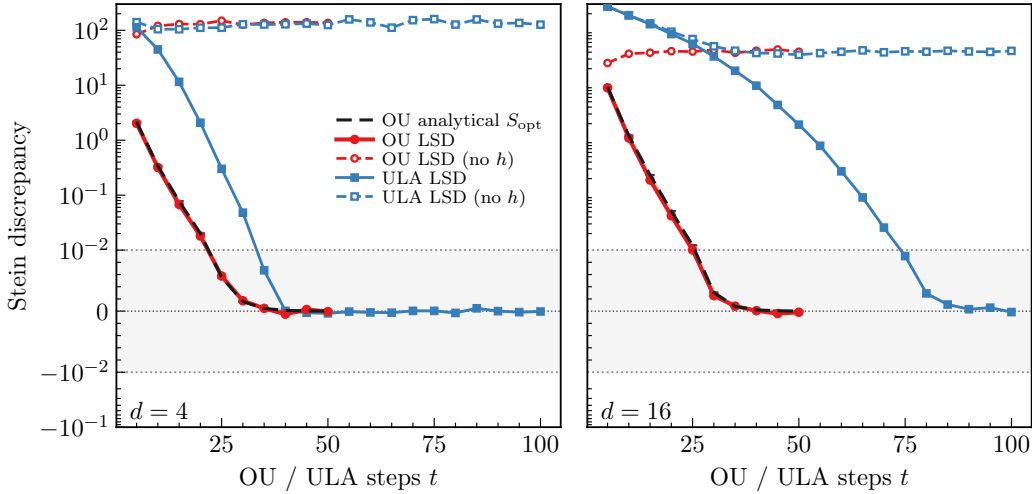


Figure 3: The LSD as a function of the number of steps for the exact OU reference process with $\eta = 0.5$ (red) and the underdamped Langevin algorithm (ULA, blue). Dashed lines show the boundary-function ablation. The black line shows the analytic optimal Stein discrepancy S_{opt} .

The validation results are shown in Fig. 3. For the exact OU samples, the learned LSD tracks the analytic value S_{opt} , supporting its use as a quantitative convergence diagnostic. The ULA curves show similar monotonic decay towards zero, with the higher-dimensional case converging more slowly, qualitatively consistent with the expected $\sqrt{d}$ scaling of Metropolis-adjusted Langevin methods [88]. In the absence of the boundary function, the remaining finite boundary contribution prevents the Stein discrepancy from approaching zero.